Example

A truck rental company has 3 locations, where you can rent moving trucks. You can return them to any other location. For simplicity, suppose that every customer returns their truck the next day. Let V_k be the vector whose entries are the number of trucks in locations 1, 2, and 3, respectively. Let A be the matrix whose (i, j)-entry is the probability that a customer renting a truck from location j returns it to location i. Consider

$$A = \begin{array}{c} \overbrace{\begin{array}{ccc} 1 & 2 & 3 \end{array}}^{\text{rented from}} \\ \begin{bmatrix} 0.3 & 0.4 & 0.5 \\ 0.3 & 0.4 & 0.3 \\ 0.4 & 0.2 & 0.2 \end{bmatrix} \end{array} \begin{array}{c} \\ \left.\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right\} \text{returned to} \end{array}$$

Assume that initially the trucks are distributed in the 3 locations as: $\longrightarrow \begin{bmatrix} 7 \\ 3 \\ 8 \end{bmatrix}$

and find out the percentages of trucks that will be distributed in each location in the long run.

$$\det(A - \lambda I) = 0$$

$$\det\begin{bmatrix} 0.3-\lambda & 0.4 & 0.5 \\ 0.3 & 0.4-\lambda & 0.3 \\ 0.4 & 0.2 & 0.2-\lambda \end{bmatrix} = \det\begin{bmatrix} -\lambda & \lambda & 0.2 \\ 0.3 & 0.4-\lambda & 0.3 \\ 0.4 & 0.2 & 0.2-\lambda \end{bmatrix} = \det\begin{bmatrix} -\lambda & \lambda & \lambda+0.2 \\ 0.3 & 0.4-\lambda & 0 \\ 0.4 & 0.2 & -(\lambda+0.2) \end{bmatrix}$$

$$= \det\begin{bmatrix} -\lambda+0.4 & \lambda+0.2 & 0 \\ 0.3 & 0.4-\lambda & 0 \\ 0.4 & 0.2 & -(\lambda+0.2) \end{bmatrix} = \det\begin{bmatrix} -\lambda+0.4 & 0.6 & 0 \\ 0.3 & 0.7-\lambda & 0 \\ 0.4 & 0.6 & -(\lambda+0.2) \end{bmatrix} = \det\begin{bmatrix} 1-\lambda & 0.6 & 0 \\ 1-\lambda & 0.7-\lambda & 0 \\ 1 & 0.6 & -(\lambda+0.2) \end{bmatrix}$$

$$= \det\begin{bmatrix} 1-\lambda & 0.6 & 0 \\ 0 & 0.1-\lambda & 0 \\ 1 & 0.6 & -(\lambda+0.2) \end{bmatrix} = -(\lambda+0.2)(1-\lambda)(0.1-\lambda) = 0$$

$$\boxed{\lambda_1 = 1\ ,\ \lambda_2 = -0.2\ ,\ \lambda_3 = 0.1}$$

$\lambda_1 = 1$ is the dominant eigenvalue.

$$\boxed{\lambda_1 = 1} \quad \left[\begin{array}{ccc|c} -0.7 & 0.4 & 0.5 & 0 \\ 0.3 & -0.6 & 0.3 & 0 \\ 0.4 & 0.2 & -0.8 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} -0.7 & 0.4 & 0.5 & 0 \\ -0.4 & -0.2 & 0.8 & 0 \\ 0.4 & 0.2 & -0.8 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -\frac{4}{7} & -\frac{5}{7} & 0 \\ 1 & \frac{1}{2} & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -\frac{4}{7} & -\frac{5}{7} & 0 \\ 0 & \frac{15}{14} & -\frac{9}{7} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\to \vec{x} = \begin{bmatrix} \frac{7}{5}t \\ \frac{6}{5}t \\ t \end{bmatrix} = \frac{t}{5}\begin{bmatrix} 7 \\ 6 \\ 5 \end{bmatrix} = S \begin{bmatrix} \frac{7}{18} \\ \frac{1}{3} \\ \frac{5}{18} \end{bmatrix} \to \vec{x}_1 = \begin{bmatrix} 7 \\ 6 \\ 5 \end{bmatrix} \text{ for } \lambda_1 = 1$$

Similarly for other eigen values we find:

$$\vec{x}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \text{ for } \lambda_2 = -0.2$$

$$\vec{x}_3 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \text{ for } \lambda_3 = 0.1$$

The set $\{\vec{x}_1, \vec{x}_2, \vec{x}_3\}$ is linearly independent

Since $\vec{x}_i$s are eigenvectors of A. Hence, we have:

$$A^k \vec{x}_0 = A^k(a_1\vec{x}_1 + a_2\vec{x}_2 + a_3\vec{x}_3) = a_1(A^k\vec{x}_1) + a_2(A^k\vec{x}_2) + a_3(A^k\vec{x}_3) = a_1\lambda_1^k\vec{x}_1 + a_2\lambda_2^k\vec{x}_2 + a_3\lambda_3^k\vec{x}_3$$

(I) when $k \to \infty$, The dominant Term is $a_1\lambda_1^k\vec{x}_1$.

(II) In This problem $\vec{x}_0 = \vec{x}_1 + \vec{x}_2 + \vec{x}_3$ which means $a_1 = 1$.

$$\to A^k\vec{x}_0 \xrightarrow{\text{when } k \to 0} (1)(1)^k \begin{bmatrix} 7 \\ 6 \\ 5 \end{bmatrix}$$

Problem

Consider an owl population of 100 adult females and 40 juvenile females. As biologists, we wish to study the growth of the population having determined the following.

1. The number of juvenile females hatched in any year is twice the number of adult females in the previous year.
2. Half the adult females in any year survive to the next year.
3. One quarter of the juvenile females in any year survive to adulthood.

Will the owls survive?

Let a_n be the number of adult females after n years and let j_n be the number of adult females after n years.

$$V_0 = \begin{bmatrix} 100 \\ 40 \end{bmatrix}$$

1. $j_{n+1} = 2a_n$

2. 3. $\left.\right\} a_{n+1} = \frac{a_n}{2} + \frac{j_n}{4}$

State vector after n years: $V_n = \begin{bmatrix} a_n \\ j_n \end{bmatrix}$

$$V_{n+1} = \begin{bmatrix} \frac{a_n}{2} + \frac{j_n}{4} \\ 2a_n \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{bmatrix} \begin{bmatrix} a_n \\ j_n \end{bmatrix} = A V_n$$

$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 2 & 0 \end{bmatrix}$ has two eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -\frac{1}{2}$.

Eigenvectors are $\vec{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{x}_2 = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$.

$\longrightarrow D = \begin{bmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \quad P = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$

$P^{-1} = \frac{1}{6}\begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}$

$$V_n = \begin{bmatrix} a_n \\ j_n \end{bmatrix} = A^n \begin{bmatrix} a_0 \\ j_0 \end{bmatrix} = PD^nP^{-1}\begin{bmatrix} a_0 \\ j_0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1^n & 0 \\ 0 & (-\frac{1}{2})^n \end{bmatrix} \frac{1}{6}\begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 100 \\ 40 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 440 + 160(-\frac{1}{2})^n \\ 880 - 640(-\frac{1}{2})^n \end{bmatrix}$$

Note that we can approximate V_n as follows:

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = P^{-1} V_0 \qquad b_1 = \frac{1}{6}\begin{bmatrix} 4 & 1 \end{bmatrix}\begin{bmatrix} 100 \\ 40 \end{bmatrix} = \frac{220}{3}$$

$$V_n \approx b_1 \lambda_1^n \vec{x}_1 = \left(\frac{220}{3}\right)(1)^n \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{220}{3} \\ \frac{440}{3} \end{bmatrix}$$ which agrees with our previous result:

$$V_n = \frac{1}{6}\begin{bmatrix} 440 + 160\left(-\frac{1}{2}\right)^n \\ 880 - 640\left(-\frac{1}{2}\right)^n \end{bmatrix}$$